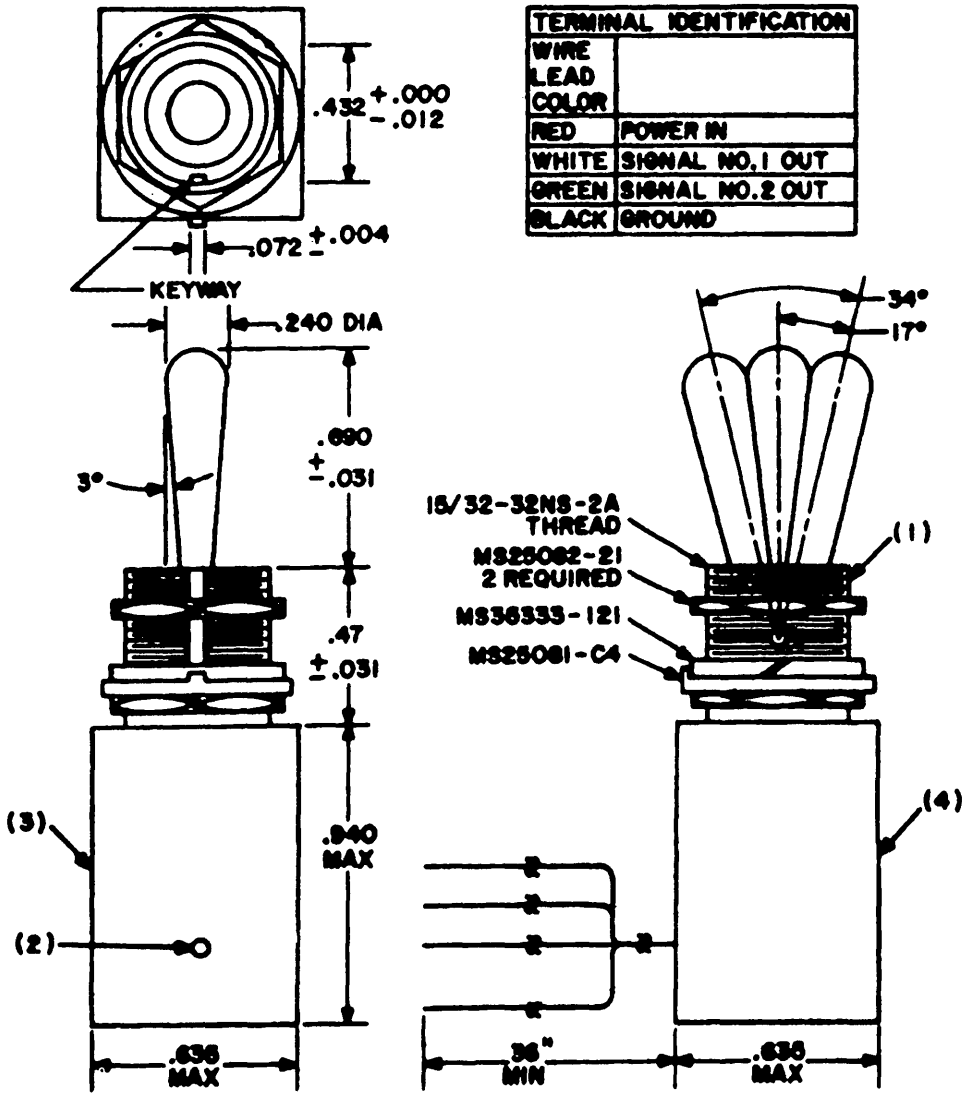
MIL-S-81619/1A
30 August 1974
Superseding
MIL-S-81619/1(AS).
8 October 1969

MILITARY SPECIFICATION SHEET

SWITCH, SOLID STATE TRANSDUCER, ELECTRONIC TRANSDUCER; TOGGLE TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch described herein shall consist of this document and the issue in effect of Specification MIL-S-81619.



TERMINAL IDENTIFICATION	
WIRE LEAD COLOR	
RED	POWER IN
WHITE	SIGNAL NO.1 OUT
GREEN	SIGNAL NO.2 OUT
BLACK	GROUND

NOTES:
1. THREAD TO WITHIN 0.062 OF SHOULDER.
2. WIRE LEADS SHALL EMERGE FROM LOWER HALF OF SWITCH BODY BELOW POSITIONING KEYWAY.
3. CONTOUR OF SWITCH BODY OPTIONAL PROVIDED MAXIMUM DIMENSIONS ARE NOT EXCEEDED.
4. IDENTIFICATION TO BE PLACED ON SURFACE OF SWITCH BODY LOCATED 180 DEGREES FROM POSITIONING KEYWAY.

FIGURE 1 SWITCH, ELECTRONIC TRANSDUCER, TOGGLE TYPE

FSC 5930

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REQUIREMENTS:

OPERATIONAL AND DIMENSIONAL

Dimensions	In inches*
Tolerances (when not specified)	±0.01 inch on decimals ±0.5° on angles
Actuator Operation	Applicable
Operating Force	0.8 pound minimum
Pretravel	Not applicable
Movement Differential	Not applicable
Overtravel	Not applicable
Terminals	Flexible wire (MIL-W-81381/1-26)
Weight	0.08 pound maximum

* Metric equivalents shall be based upon 1 inch = 25.40 millimeters.

MECHANICAL

Strength of Actuating Means	25.0 pounds
Strength of Mounting Means	Applicable
Terminal Strength	5.0 pounds (tensile)
Resistance to Soldering Heat	Not applicable
Solderability	Not applicable
Enclosure Type	Watertight
Operating Cycles	150,000 minimum

ELECTRICAL

Dash No.	Logical State of Output Signals With Toggle Lever in These Positions						Input Power Watts - Maximum	
	Opposite Keyway Side		Center		Keyway Side			
	Signal No.		Signal No.		Signal No.		Output	State
	1	2	1	2	1	2	1	0
001	0	1	0	0	1	0	0.35	0.20
002	1	None	None		0	None	0.25	0.10
004	0	1	0	0	Momentary 1	0	0.35	0.20
005	1	None	0	None	MOM 1	None		
006	0	None	None		MOM 1	None	0.25	0.10

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Absolute Supply Voltage	8 volts dc maximum
Absolute Supply Current	50 ma maximum
Supply Voltage	6 volts ± 5% dc
Output Voltage and Current	
Logical '1'	4.0 volts dc minimum
	9 ma minimum
Logical '0'	0.6 volts dc maximum
	7 ma minimum (sinking)
Load Impedance	500 ± 10% ohms, resistive
Output: Rise Time	10 microseconds maximum
Fall Time	10 microseconds maximum
Output Voltage Waveform	Applicable
Short Circuit	Applicable
Transient Voltage	Applicable
Insulation Resistance	Applicable
Dielectric Withstanding Voltage	Applicable

ENVIRONMENTAL

Temperature	
Operating	-54° C to +95° C
Storage	-54° C to +150° C
Shock	
Mechanical	50 g for 11 milliseconds
Temperature	-54° C and +71° C
Vibration	Applicable
Acceleration	50 g
Salt Fog	Applicable
Humidity	Applicable
Sand and Dust	Applicable
Electromagnetic Compatibility	Applicable
Magnetic Compatibility	Applicable
Temperature-Altitude	Applicable
Operating Ambient	
Temperature	-54° C to +95° C
Altitude	Sea level to 100,000 ft.

Custodians:
Navy - AS
Air Force - 80

Preparing Activity:
Navy - AS

Project No. 5930-0713-01

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during steps 1 and 3 at a rate of n ready actuated switches shall be observed

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL	OMB Approval No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-S-81619/1A SWITCH, SOLID STATE TRANSDUCER, ELECTRONIC TRANSDUCER; TOGGLE TYPE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)	TELEPHONE NO.
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